

MD. AMIT HASAN

Oceanography Researcher | GIS & Remote Sensing | Data Analysis & Modelling | GeoAI/XAI
Dhaka, Bangladesh | +8801612977222 | ocn19011.amit@bmu.edu.bd | amithasan.ocean@gmail.com |

Portfolio: <https://amithasanbd.github.io/> | [ResearchGate](#) | [LinkedIn](#) | [Google Scholar](#)



PROFESSIONAL PROFILE

Oceanography researcher with 3+ years of part-time and project-based experience across climate, coastal, marine, water-quality, and environmental research. Skilled in research design, literature synthesis, field and laboratory-supported data collection, GIS/remote sensing, statistical analysis, scientific visualization, technical reporting, and manuscript development. Proficient in Python, R, SPSS, Excel, ArcGIS Pro, QGIS, Google Earth Engine, and OriginPro, with working exposure to numerical modelling, GeoAI, explainable AI, and data-driven environmental applications. Currently pursuing an M.Sc. in Oceanography at Bangladesh Maritime University and interested in interdisciplinary research linking ocean-climate processes, geospatial intelligence, ecosystem resilience, and decision support.

EDUCATION

M.Sc. in Oceanography	Expected Dec 2026
<i>Bangladesh Maritime University</i> Thesis ongoing	
• Thesis Title: Developing JeleBondhu: A GeoAI-Driven Digital Twin Ocean Framework for Climate-Smart Sustainable Fishing Advisory and Forecasting in the Bay of Bengal	
B.Sc. in Oceanography	2025
<i>Bangladesh Maritime University</i> CGPA: 3.55/4.00	
• Thesis Titled: Identification of Potential Marine Protected Areas in the Coastal Region of Bangladesh: A Multi-Criteria Decision Making (MCDM) Approach Bay of Bengal	

CORE COMPETENCIES

Research & Scientific Writing	Research design; literature review; evidence synthesis; technical reports; manuscript preparation; figures and tables.
GIS & Remote Sensing	ArcGIS Pro; QGIS; Google Earth Engine; SeaDAS; raster/vector processing; satellite analysis; spatial mapping; field validation.
Data Analysis & Computing	Python; R; SPSS; MS Excel; Power BI; OriginPro; data cleaning; QA/QC; statistics; visualization; interpretation.
Modelling & AI	Basic Delft3D, HEC-RAS, CROCO/OFS concepts; GeoAI; explainable AI (XAI); machine-learning and forecasting workflows.
Field & Laboratory	Environmental/oceanographic sampling; CTD; multiparameter water-quality meter; Algal Torch; spectrophotometer; AAS exposure.
Project Coordination	M&E inputs; deliverable tracking; field planning; stakeholder communication; documentation; workshops and research dissemination.

PROFESSIONAL EXPERIENCE

Project Coordinator	Aug 2026 - Present
<i>Youth4Climate Mangrove Restoration and Monitoring Project</i> Contractual	
• Coordinate field assessment, restoration planning, stakeholder engagement, project documentation, M&E inputs, procurement records, and progress tracking.	

- Support GIS/RS-based site prioritization and risk assessment using satellite data, Python, and Google Earth Engine.

Research Coordinator

Jul 2026 - Present

Hydro-Climate and Ocean Centre (HCO) | Part-Time

- Coordinated hydro-climate, water-resource, environmental, and GIS/RS research activities; supported data processing, modelling, technical reporting, and project development.

Research Associate

Oct 2024 - Aug 2026

Bangladesh Maritime University | Part-Time

- Supported climate, coastal, water-quality, and environmental research through field/sample data collection, quantitative analysis, GIS mapping, interpretation, and technical reporting.
- Prepared publication-ready maps, figures, tables, presentations, and manuscript materials using Python, R, SPSS, Excel, GIS/RS, and visualization workflows.

Assistant Research Officer / Data Analyst

Dec 2024 - Apr 2026

DSD Research Firm | Part-Time / Contractual

- Supported proposal preparation, field coordination, questionnaire-based data collection, data cleaning/validation, statistical analysis, reporting, and deliverable management.

Researcher

Jan 2021 - Dec 2024

National Oceanographic and Maritime Institute (NOAMI) | Part-Time

- Supported oceanographic and environmental studies through fieldwork, instrument support, dataset processing, sample/data documentation, analysis, and technical reporting.

RESEARCH PROJECT EXPERIENCE

- **Youth4Climate Mangrove Restoration and Monitoring Project** — Project coordination; GIS/RS site prioritization; ground-truthing; restoration planning; M&E and stakeholder coordination.
- **Climate-Induced Shifts in Potential Fishing Zones** — Fisheries, blue economy, and coastal livelihood resilience analysis using satellite-derived oceanographic data.
- **Existing and Potential Marine Protected Areas of Bangladesh** — Biodiversity conservation, area-based management, spatial assessment, and sustainable marine-management planning.
- **Sundarbans Mangrove Ecosystem Health Assessment** — Ecosystem condition, resilience, conservation concerns, and climate-vulnerability context.
- **Dhaka (Shitalakshya) River Ecosystem Health Assessment** — Water quality, ecological stress, and river-health dynamics for management decisions.
- **Marine Heatwaves and Coastal Bangladesh** — Thermal-stress patterns and ecological implications of climate-driven ocean stress.
- **Endangered Species Assessment Using eDNA** — Conservation-focused research using environmental DNA for aquatic species detection and monitoring.
- **Fishnet Detection GeoAI** - Ongoing data-driven research applying geospatial AI and environmental modelling to marine spatial and biogeochemical questions.

SELECTED PEER-REVIEWED PUBLICATIONS

1. Hasan, M. A., et al. (2026). Assessing the dynamics of water quality and implications on bio-chemical health of river ecosystem: A RS-based study of the Shitalakshya River, Dhaka. *Bangladesh Maritime Journal*, 10(1), 313-329. DOI: 10.70279/bmj-v2-1124.
2. Sowrav, S. F. F., et al., and Hasan, M. A. (2026). Water quality affects socioeconomic traits of climate-vulnerable coastal people: case study on Dacope and Mongla. *Journal of Integrated Disaster Risk Management*, 16(1). DOI: 10.5595/001c.158653.

Additional manuscripts/research in progress: PFZ variability; marine heatwaves; OMZ modelling; mangrove restoration; marine spatial planning; air-quality assessment.

SELECTED TRAINING & CERTIFICATIONS

- **Technical Capacity on Water-Level Data Collection, Processing & Climate Hazard Analysis**
— Embassy of France in Bangladesh, ICOG & University of Dhaka, Sep 2026
- **Climate Change, & Research Methodology**
— CCDB, Dec 2025
- **Marine Megafauna and Fisheries Survey Methodology**
— WCS, 2025
- **Workshop on Coastal & Regional Ocean Community Model (CROCO)** — BMU, Aug 2025
- **Earth Observation for Enhanced Fisheries MCS**
— UNESCO/IOC, Jul 2025
- **Numerical Model & AI for Ocean Science (NMAI4OS)**
— NOAMI, 2024
- **Radar Meteorology Applied Lecture Series**
— SAMA/ACARR-CUSAT, 2024 (Outstanding)
- **Satellite Meteorology Lecture Series**
— SAMA/BIT Mesra, 2023–2024
- **SeaDAS for Optical Remote Sensing & Water Quality Monitoring** — NASA ARSET, Feb 2024
- **Satellite Meteorology Lecture Series**
— SAMA/BIT Mesra, 2023–2024
- **Ocean Model for OFS Implementation**
— UNESCO/IOC & BMKG, Jul 2023
- **Building Research Career for Early-Career Professionals & Students** — Nirmol BD, 2023
- **Data Science & Machine Learning with Python** — NOAMI & BCA, 2021

AWARDS & LEADERSHIP

Awards: Climate Change Scholarship - CCDB (2025); 3rd Place - NOAMI Ocean Olympiad (2025); Best Trainee - Radar Meteorology Lecture Series (2024); Best Trainee - Data Science with Python (2022).

Leadership: Founding Team Head, BMU Ocean Club (2024-2025); Advisor, BMU Ocean Club (2026-Present).

ADDITIONAL INFORMATION

Languages: Bangla - Native/Fluent; English - Professional Working Proficiency.

Research Interests: Ocean-climate dynamics; coastal resilience; remote sensing/GIS; GeoAI/XAI; environmental data science; marine ecosystem and water-quality assessment.

REFERENCES

- Dr. S. M. Mustafizur Rahman** — Associate Professor, Bangladesh Maritime University | mustafizur.ocn@bmu.edu.bd | +8801401807894
- Dr. Mohan Kumar Das** — Executive Director, Hydro-Climate & Ocean Center (HCO) | mohan.smrc@gmail.com | +8801913630710
- Sheikh Fahim Faysal Sowrav_** — Lecturer, Bangladesh Maritime University | sowrav.ocn@bmu.edu.bd | +8801986761741